

Kevin, 24 · stab wound to mid-back at T8 level · awake, breathing on his own · **right leg weakness with hyperreflexia** + **loss of pain and temperature on left leg** · preserved vibration/proprioception right · **Brown-Séquard syndrome**

BIO 004 · WEEK 8 · CASE DEEP DIVE

The Spinal Cord

A stab wound to the back. Motor loss on one side and pain loss on the other. A textbook diagnosis from anatomy alone.

PATIENT CASE

Kevin is 24. He was stabbed in the mid-back during an altercation an hour ago. The knife entered just to the right of his spine at roughly the **T8 level**. He is awake, alert, hemodynamically stable, breathing on his own.

On neurologic exam: **weakness of the right leg with hyperreflexia and a positive Babinski; preserved vibration and proprioception on the right; and loss of pain and temperature sensation on the left leg, starting a few segments below the lesion.** He has dissociated sensory loss — exactly the pattern of **Brown-Séquard syndrome**, a hemisection of the spinal cord.

Kevin's findings are anatomy in action. Three tracts cross at different levels; a half-cord lesion separates their effects across the two sides of the body. To read his exam, you need the spinal cord's gross structure, its gray and white matter organization, and the three tracts that matter most clinically.

GUIDING QUESTION

How is the spinal cord built, what runs up and down inside it, and how does a hemisection split motor, fine touch, and pain/temperature loss between the two sides of the body?

From foramen magnum to L1-L2

The spinal cord extends from the foramen magnum to about **L1-L2 in adults** (lower in children). It is shorter than the vertebral column because the column grows faster after birth. Below L1-L2, the lumbar and sacral nerve roots float freely in CSF as the **cauda equina** ("horse's tail") — which is why lumbar puncture below L2 is safe.

Two enlargements: **cervical enlargement** (C4-T1) supplies the upper limbs (brachial plexus); **lumbosacral enlargement** (L1-S3) supplies the lower limbs (lumbar and sacral plexuses).

The cord tapers to a point called the **conus medullaris** at ~L1. A thin strand (*filum terminale*) anchors it to the coccyx.

H-shaped gray in the middle, white outside

Unlike the brain (gray outside), the spinal cord has **gray matter in the center** (an H-shaped or butterfly arrangement) and **white matter on the outside**.



Dorsal (posterior) horns

Sensory. Receive incoming signals from sensory neurons.



Ventral (anterior) horns

Motor. Contain lower motor neurons whose axons exit to muscles.



Lateral horns (T1-L2)

Sympathetic preganglionic neurons.



White matter columns

Dorsal, lateral, ventral funiculi. Carry ascending and descending tracts.

The clinically important highways

TRACT	CARRIES	CROSSES WHERE
Dorsal column (medial lemniscus)	fine touch, vibration, proprioception	in the MEDULLA (above the cord)
Spinothalamic	pain, temperature, crude touch	at level of ENTRY into cord (crosses within 1-2 segments)
Lateral corticospinal	voluntary motor	in the MEDULLA (pyramidal decussation)

Why Brown-Séquard splits the loss. A right-sided cord lesion at T8 cuts: (1) Right dorsal column → loss of fine touch/vibration/proprioception on the RIGHT (ipsilateral) below the lesion (because dorsal columns haven't crossed yet at this level). (2) Right corticospinal tract → upper motor neuron weakness on the RIGHT below the lesion (because corticospinal already crossed in the medulla, so the right tract is now serving the right body). (3) Right spinothalamic → loss of pain/temperature on the LEFT (contralateral) starting a few segments below the lesion (because spinothalamic fibers entered on the left, crossed within 1-2 segments, and are now ascending on the right where they got cut).

31 pairs exiting through intervertebral foramina

The spinal cord gives off 31 pairs of **spinal nerves**: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, 1 coccygeal. Each spinal nerve is formed by the union of a **dorsal (sensory) root** and a **ventral (motor) root** just inside the intervertebral foramen.

Each spinal nerve splits into a **dorsal ramus** (innervates back muscles and skin over the back) and a **ventral ramus** (everything else; forms plexuses in the limb regions). Covered on the Plexuses and PNS pages.

Bedside maps

Each spinal nerve's sensory fibers innervate a strip of skin called a **dermatome**. Memorizing a handful is high-yield: *C4 shoulder, C6 thumb, C7 middle finger, C8 pinky, T4 nipples, T10 umbilicus, L4 medial calf/knee, L5 great toe, S1 pinky toe/lateral foot, S2-S4 perineum*.

Spinal reflexes test specific levels: *biceps reflex C5-C6, brachioradialis C5-C6, triceps C7-C8, patellar L2-L4, Achilles S1-S2*.

Same three layers as the brain

Three meninges wrap the cord: dura, arachnoid, pia. **Epidural space** here actually contains fat and a venous plexus (where epidural injections go). **Subarachnoid space** contains CSF and extends below the cord around the cauda equina.



DRAW ZONE 1 -- Sketch a spinal cord cross-section. Label dorsal/ventral horns, lateral horns, dorsal columns, lateral corticospinal tract, spinothalamic tract.

Mark each tract's function and where it crosses. Then shade what Kevin's right-sided lesion destroys.



DRAW ZONE 2 -- Diagram Kevin's sensory and motor findings on a body figure: right motor loss + ipsilateral fine touch loss; contralateral (left) pain/temperature loss.

Color each tract's territory with a different color so the dissociation is obvious.

RETURNING TO KEVIN

One side cut, two sides lose

Brown-Séquard syndrome is rare in practice but a textbook test of spinal cord tract anatomy. Every finding is predictable from where each tract crosses.

"Right leg weakness with hyperreflexia"

Right lateral corticospinal tract damaged. UMN signs (hyperreflexia, Babinski) appear because the tract already crossed in the medulla.

"Preserved vibration/proprioception RIGHT"

Wait — Brown-Séquard actually **LOSES** dorsal column function on the ipsilateral side. Kevin's preserved vibration/proprioception on the right suggests partial sparing. Classic Brown-Séquard loses ipsilateral fine touch.

"Loss of pain/temperature on LEFT"

Spinothalamic fibers from the left leg entered on the left, crossed within 1-2 segments, ascended on the right — and got cut at T8. Result: left-sided pain/temp loss starting a couple of segments below T8.

"Why the dissociation"

Different tracts cross at different levels. A unilateral cord lesion separates the motor and dorsal column losses (ipsilateral) from the pain/temperature loss (contralateral).

"Prognosis"

Brown-Séquard has the best prognosis of incomplete cord syndromes. ~90% of patients regain bowel/bladder function; most regain ambulation.

"Why his breathing is intact"

Phrenic nerve to the diaphragm comes from C3-C5. T8 lesions spare diaphragmatic function. Intercostal weakness may impair cough.

Tract, crossing, dorsal, ventral. Kevin's exam is the spinal cord teaching you tract anatomy in real time.

